

Jose Sanchez Gonzalez

AI Engineer / RL Environments and Agent Evaluation

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Summary

M.S. Artificial Intelligence developing robot-policy evaluation in Isaac Lab, Isaac Sim, and MuJoCo. Combines controlled sim-to-sim comparisons, imitation-learning failure analysis, and Quest VR integration on physical humanoid arms.

Technical Skills

Languages and Learning: Python, C++, PyTorch, NumPy, PPO, imitation learning

Simulation: Isaac Lab, Isaac Sim, MuJoCo, PhysX cloth, OpenUSD

Robot Integration: Inverse kinematics, SocketCAN, WebXR, gravity feedforward, ROS 2

Evaluation and Systems: Matched controls, policy export, ONNX, pytest, Docker, Linux, Slurm

Projects

Humanoid Robustness Ladder | Isaac Lab, MuJoCo, PPO, ONNX, Python

- Automated PPO export and MuJoCo sim-to-sim scoring through the upstream Berkeley Humanoid Lite controller, with matched controls and scenario-linked artifacts.
- Compared five randomization settings across 450 MuJoCo episodes: 23.3% falls without randomization versus 0% for the default (90 episodes per setting).

Bimanual Garment Folding Evaluation | SmolVLA, PyTorch, Isaac Sim, PhysX cloth

- Integrated SmolVLA with PhysX cloth, three camera views, and the LeHome checker; separated policy rollouts from demonstration replay.
- Evaluated recovery-supervised fine-tunes with paired, source-pinned screens; short-horizon gains missed the long-horizon gate, retaining the baseline.

Berkeley Humanoid Lite: Quest VR Teleoperation | Quest 2, WebXR, Python, inverse kinematics, SocketCAN

- Built Quest 2/WebXR teleoperation for simulated and physical Berkeley Humanoid Lite arms: Python/UDP pose bridge, inverse kinematics, and SocketCAN actuation.
- Integrated gravity feedforward, grip-based motion gating, and headset calibration; added camera, relative-depth/optical-flow, lidar, and IMU views.

MetaNavIT: Retrieval Evaluation and APIs | Python, PostgreSQL, pgvector, BM25, RRF, BGE

- Contributed retrieval/data APIs on a six-person team; extended the fork with hybrid retrieval, BGE reranking, and PostgreSQL/pgvector parity checks.

Agentic Match Forecasting | XGBoost, LangGraph, MCP, FastAPI, Docker

- Connected XGBoost forecasts and contextual MCP tools through LangGraph/FastAPI, with calibration, conformal sets, and approval before staked recommendations.

Experience

Graduate Teaching Assistant, Oregon State University | Sep 2025 – Jun 2026

- Led 2-hour Operating Systems (C++) and Computational Methods (COMSOL) studios for 25+ students; taught debugging, concurrency, and numerical simulation.

Math and Computer Science Tutor, Oregon State University | Nov 2022 – Jun 2025

- Tutored 50+ students in mathematics and computer science; adapted debugging guidance and technical explanations to individual learning needs.

Education

Oregon State University, Corvallis, OR

Master of Science in Artificial Intelligence | Jun 2026

Bachelor of Science in Computer Science (Applied: Data Science); Mathematics Minor | Jun 2025